

Gentle Introduction to Computation Methods for Drug Design

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Drug discovery is an arduous journey with many unknowns and traps throughout the path that lead to failure. To perform a large number of trials iteratively, computer simulations are convenient enabling tens of thousands of calculations. The sheer number of calculations can provide crucial insights, and possibly, rule out dead ends very early in the discovery cycle. Drug discovery is a multi-faceted science that requires a great deal of art and innovative thinking and often luck plays a good role. For the latter factor, we have no control, however, the best we can do is to base our science on solid foundation. In computational chemistry and biology, we often ask an important question- how do we train computers to think like humans? However, this is not a worry because we do not need to train computers to think like humans. As a matter of fact, computers are extremely fast but dumb machines. Whereas humans are extremely slow, lazy but smart thinkers. Now, the division of labour between humans and computers should be straightforward; let the computers do the repetitive tasks of solving several complex mathematical equations that can be programmed to describe chemistry and biology. This is done by employing the laws of physics that govern the movement of the atoms, and therefore molecules. Humans can engross themselves in decision-making that is beyond mathematical equations. In the recent times with generative artificial intelligence (**GenAI**), computers can perform some level of thinking. However, their chemical and biological validity, in a general sense, remains to be tested. GenAI, therefore, deserves an appreciation to at least recognise the idea that computers could be made to think, not in a biological sense, rather in a mathematical sense that can be converted to chemical or biological models. However, the latter part has been extremely challenging and yet to be conquered.

This chapter can be treated as a short summary on topics that will be covered in great detail in the chapters to follow. Note that this book is intended to be a practical guide, therefore, we will emphasise more the practical aspects of molecular one needs to know before use it for drug design and discovery. However, we have made efforts to cover the theoretical foundations on topics that we deemed crucial to successfully apply molecular docking.

1 Historical Background

1.1 Early Foundations

The conceptual foundations of structure-based drug design can be traced back to the early 20th century with Emil Fischer's lock-and-key theory proposed in the late 1800s.

This theory postulated that ligands fit into receptors like a key into a lock, representing the first mechanistic understanding of molecular recognition. This concept was later refined by Daniel Koshland's induced-fit theory in 1958, which recognized that proteins are dynamic entities that undergo conformational changes upon ligand binding. The induced-fit theory suggested that the active site of the protein is continually reshaped by interactions with the ligands as they interact with the protein, providing a more accurate description of binding events than the rigid treatment.

1.2 The Birth of the Protein Data Bank

A pivotal moment in the history of SBDD came in 1971 with the establishment of the Protein Data Bank as the first open-access, digital-data resource in biology. The PDB was created to gather structural biologists at a single platform and has evolved from being a collaborative network to a full-fledged database with an extensive list of protein structures, nucleic acid-containing structures, and computational models. The availability of three-dimensional atomic-level structures of biomolecules determined using X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy, and electron microscopy fundamentally transformed drug discovery.

1.3 The X-ray Crystallography Revolution

X-ray crystallography emerged as the cornerstone technique for determining protein structures at atomic resolution. The method involves crystallization of biological molecules, diffraction of X-rays by crystals, and data collection and analysis. X-ray crystallography has been instrumental in revealing the three-dimensional architecture of drug targets, enabling rational drug design approaches. The impact of X-ray crystallography on drug discovery and development has been transformative, providing crucial structural insights into the interactions between therapeutic compounds and their target proteins.

1.4 Evolution of Computational Methods

The earliest reported docking methods were based on the lock-and-key theory, treating both the ligand and receptor as rigid bodies. As computational capabilities advanced, docking evolved to incorporate ligand flexibility while keeping the receptor rigid, which remained the most popular method for an extended period. Recently, many efforts have been made to address receptor flexibility, although flexible receptor docking, especially backbone flexibility in receptors, still presents a major challenge for available docking methods. The development of sophisticated algorithms and scoring functions has progressively improved the accuracy and efficiency of molecular docking simulations.

2 Molecular Recognition and Binding

Understanding protein-ligand interactions at the molecular level is central to studying how small molecules interact with their biological receptor(s) to show their therapeutic effect. Thus binding phenomenon may involve, but not limited to, multiple types of non-covalent interactions such as hydrogen bonds, electrostatic interactions, van der Waals forces, and hydrophobic interactions. The binding affinity between a ligand and its receptor can be determined by the sum of favourable and unfavourable interactions, with the overall

binding free energy dictating the strength of the complex. Protein-ligand binding affinity prediction quantifies the binding strength, providing crucial insights for lead optimisation by changing functional groups that lead to maximum gains.

The binding affinity is determined experimentally in the form of inhibition constants (K_i), dissociation constants (K_d), or half-maximal inhibitory concentrations (IC_{50}). If one is using isothermal calorimetry, then direct measurements to thermodynamic qualities like the change in enthalpy due to ligand binding (ΔH), and the change in entropy ($T\Delta S$) can be made. However, experimental methods are expensive and time-consuming. Thus, efficient computational methods are capable of making accurate protein-ligand binding affinity predictions. Such accurate and high-quality computational methods can reduce the number expensive experiments by filtering unwanted molecules. The downside of computational methods are they dependent the parameters using for computing. The potential energy function that is used to represent the chemical system. The aim of this book is to ensure that the reader, perhaps, a beginner is aware of many assumptions that enable quick calculations. Also, thoroughly understand the limitations of these methods and to use them to perform calculations that they are meant or developed for. There are certain critical assumptions that ensure the speed of the calculations are, neglecting the solvation effect, and neglecting the entropy part of the free energy of binding. These, and more limitations will discussed in great detail in this later chapters of this book. It must be noted that such assumptions are somewhat important to ensure that the calculations are fast enough to finish several hundreds or thousands docking calculations per day, as needed for virtually screening huge databases that comprise few million molecules.

3 Structure-based drug design

Structure-based drug design (SBDD) uses the knowledge of the receptor structure and the binding site thereof to understand the shape of ligands and the type of interactions required for binding. it involves the use of three-dimensional structures of biological macromolecules, such as proteins or nucleic acids, to guide the design of molecules that can bind to the target protein and modulate its activity. The lock-and-key hypothesis is the main assumption for developing software tools. Therefore, the quality of the results depend on two things assuming the software used is accurate (more on this in later chapters explaining why this is not always true):

- Quality of the receptor structure
- Confidence with which the binding site information is known

The objective of molecular docking is to enables us to predict the binding modes, and thus, affinities of small molecules to their target receptor. It models the interaction between the atoms of small molecule and receptor. Generally speaking, molecular docking is possible between any two (or more) chemical or biological entities, such as protein-protein, protein-DNA, protein-RNA and all such combinations that are biologically observed. All aforementioned binding and recognition follows the same rules of interactions governed by laws of physics. However, computationally, different potential energy functions are needed to model these binding with accuracy. We mainly focus on small molecule protein docking, nevertheless, these discussion should be applicable to other kind of docking.

3.1 Fundamental Principles of Structure-Based Drug Design

3.1.1 The SBDD Paradigm

Structure-based drug design is fundamentally dependent on knowledge of the three-dimensional structure of the biological target. The iterative process of SBDD begins with the selection of a potential target, followed by ligand identification. The target may be a macromolecular protein or enzyme involved in biosynthetic pathways. After selection of the target, extraction, purification, and determination of the structure of the protein are crucial processes. The 3D structure of the protein is developed using sophisticated techniques including X-ray crystallography, nuclear magnetic resonance spectroscopy, and cryo-electron microscopy. Once the structure of the target is identified, it is necessary to investigate the electrostatic properties of the binding site such as presence of cavities, clefts, amino acid sequences, and allosteric pockets. Using various available computer algorithms, drug-like molecules from large databases of small molecules or fragments of compounds can be docked into the binding cavities of the target protein.

3.2 The SBDD Workflow

Figure 1 shows a typical workflow for computational drug discovery. Depending on the chemical and biological information available, one might proceed by the structure-based approach or ligand-based.

3.2.1 Druggability and Target Selection

Druggability refers to the likelihood that a biological target can be modulated by a small molecule or biological drug. The first step in SBDD is the choice of an appropriate target, followed by evaluation of the structure of that target. Targets must possess characteristics that make them amenable to small molecule binding, including accessible binding sites with suitable geometry and chemical properties. Artificial intelligence models, particularly supervised machine learning algorithms, can now be trained on datasets to differentiate known druggable and non-druggable targets based on structural properties of the protein, its function, interaction partners, and its role in disease pathways.

3.2.2 Determination and Preparation

The SBDD workflow begins with obtaining the three-dimensional structure of the target protein. Structures can be obtained from the Protein Data Bank, which currently holds over 155,000 atomic-level 3D structures of biomolecules. When experimental structures are not available, protein modeling including homology modeling and *ab initio* techniques allows for creation of three-dimensional protein structures. Selection of appropriate protein structure from the PDB is a critical step, as multiple crystal structures are often available for the same protein. A systematic approach for selecting the right type of protein structure must consider factors such as resolution, completeness, presence of ligands, and biologically relevant conformations. Recent advances include AlphaFold, which has revolutionized life sciences by predicting over 200 million protein structures available in the AlphaFold Database, far exceeding the number of experimentally resolved structures.

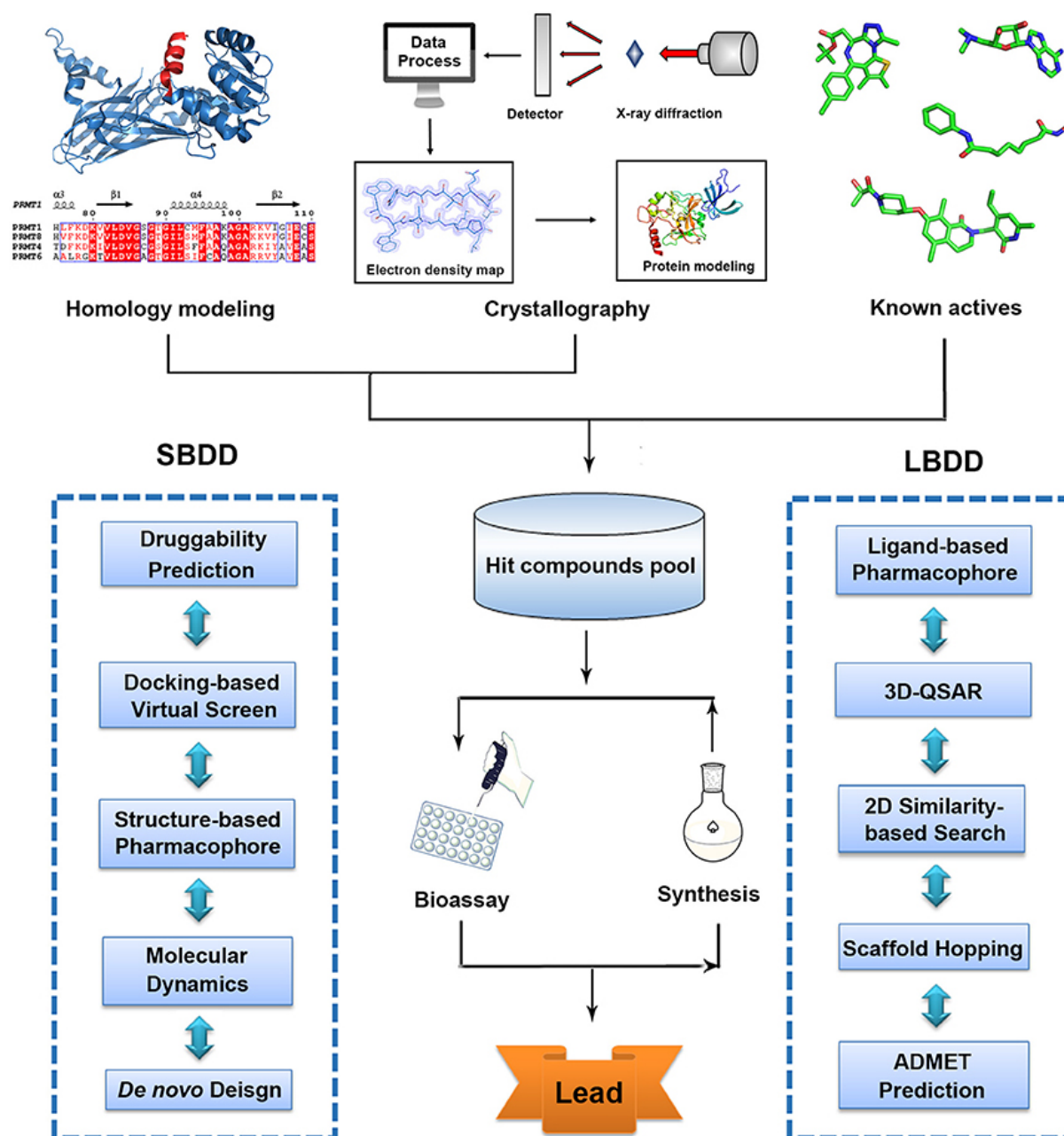


Figure 1: A typical computational workflow for drug design. This figure also compares an alternative workflow called ligand-based drug design (LBDD), which is used when receptor information is not available. This image is reproduced from 10.3389/fchem.2018.00057 under the Creative Commons Attribution License (CC BY) licence.

3.2.3 Ligand Library Preparation

After target structure preparation, the next step involves preparing libraries of compounds for virtual screening. Ligand preparation includes generating three-dimensional coordinates, assigning proper protonation states, and optimizing geometries. Compound libraries can range from commercially available compounds to virtual libraries representing chemical space that could be synthesized.

3.2.4 Virtual Screening

Virtual screening is a computational technique used to search large libraries of compounds and identify those most likely to bind to a drug target. Structure-based virtual screening uses molecular docking to test large libraries of compounds for potential binding to target proteins. Computer-aided drug design allows many molecules to be analyzed in a short time and enables simulation and prediction of several essential factors. The virtual screening process typically involves multiple stages of filtering and refinement. Initial rapid screening of large compound collections using fast scoring methods is followed by more detailed analysis of top-ranked compounds using more sophisticated scoring approaches. Consensus virtual screening between multiple docking programs can increase reliability of results. Hit Validation and Optimization Top-scoring molecules from virtual screening with high affinity toward target protein are synthesized and tested in in vitro biochemical assays. Ligands with desired therapeutic activities are further evaluated for efficacy, affinity, and potency studies. Structural insights into the complex of ligand-protein assist the analysis of variety of binding conformations and interactions. The availability of 3D structure of the target protein in complex with promising ligands offers detailed evidence of intermolecular features aiding in molecular identification process and binding pattern of the ligand. This iterative process of structure determination, docking, synthesis, and testing continues until optimized lead compounds are identified.

3.2.5 ADMET Prediction

ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) predictions refer to computational estimation of pharmacokinetic and toxicological properties of compounds. These predictions are crucial for evaluating drug-likeness of molecules and their potential success as therapeutic agents. ADMET prediction allows for early identification of potential issues with a compound’s pharmacokinetics and toxicity, reducing the risk of late-stage failure. By predicting ADMET properties computationally, resources can be focused on the most promising compounds, saving time and reducing costs associated with experimental testing. Poor ADMET properties are a major cause of drug failure in clinical trials, leading to increased costs and delayed development timelines. By identifying and addressing ADMET liabilities early in the drug discovery process, the risk of clinical failure due to pharmacokinetic or safety issues can be significantly reduced.

4 Applications and Success Stories

4.1 The HIV Protease Success Story

One of the most celebrated examples of early SBDD success came in the 1990s with the development of HIV protease inhibitors. The discovery of HIV protease inhibitors represents a major success of structure-based drug design and is regarded as a landmark achievement in pharmaceutical research. Protease inhibitors were designed to mimic the transition state of the protease’s actual substrates, with a peptide linkage replaced by a hydroxyethylene group that the protease is unable to cleave. HIV protease inhibitors fit the active site of the HIV aspartic protease and were rationally designed utilizing knowledge of the aspartyl protease’s mode of action. The most promising transition state mimic was hydroxyethylamine, which led to the discovery of saquinavir, the first protease

inhibitor. Structure-based design and synthesis of novel HIV protease inhibitors demonstrated that inhibitors could be designed specifically to interact with the backbone of HIV protease active site to combat drug resistance. X-ray crystal structures of inhibitor-bound HIV-1 protease revealed extensive interactions in the HIV protease active site including strong hydrogen bonding interactions with the backbone. This design strategy proved remarkably successful, with HIV protease inhibitors becoming highly effective against HIV and revolutionizing HIV treatment.

4.2 Oncology

Structure-based drug design has made profound contributions to cancer drug discovery. Open access to three-dimensional atomic-level biostructure information from the Protein Data Bank facilitated discovery and development of 100% of the 34 new low molecular weight, protein-targeted, antineoplastic agents approved by the US FDA 2019-2023. The impact of structural biologists and public-domain 3D biostructure data was broad and substantial, ranging from understanding target biology (100% of all drug targets), to identifying targets as likely druggable (100% of all targets), to structure-guided drug discovery (over 80% of all new small-molecule drugs). Notable examples include sotorasib (Lumakras), a covalently-acting G12C KRAS inhibitor; asciminib (Scemblix), a non-ATP-competitive inhibitor of the BCR-Abl fusion protein; and tazemostat (Tazverik), an EZH2 methyltransferase inhibitor. These success stories highlight the transformative impact of SBDD on developing therapies for previously undruggable targets.

4.3 Infectious Disease

SBDD has proven invaluable in combating infectious diseases. The availability of high-resolution structures of SARS-CoV-2 proteins enabled researchers to identify inhibitors and design vaccines rapidly. Structures of the SARS-CoV-2 main protease and spike protein have been pivotal in developing antiviral drugs and mRNA vaccines. More than 800 SARS-CoV-2 protein structures have been deposited with the Protein Data Bank, providing insights into viral transmission and revealing novel therapeutic targets. For malaria research, structure and ligand-based drug design methods including molecular docking, molecular dynamics simulations, pharmacophore modeling, QSAR, and virtual screening are widely used to design novel antimalarial agents. These modeling methods speed up antimalarial drug discovery and selection of better drug candidates for synthesis.

4.4 Neurological Disorders

Molecular docking has contributed to discovery of novel inhibitors for treating Alzheimer’s disease. Researchers used PDB structures of acetylcholinesterase to identify novel inhibitors through virtual screening. Recent advances in molecular docking techniques for targeting protein misfolding in neurodegenerative diseases have shown promise for Alzheimer’s, Parkinson’s, and Huntington’s disease. These include compounds that inhibit amyloid- β aggregation in Alzheimer’s disease, α -synuclein oligomerization in Parkinson’s disease, and mutant huntingtin aggregation in Huntington’s disease.

4.5 Ion Channels and Membrane Proteins

Ion channels play important roles in fundamental biological processes including electric signaling, muscle contraction, hormone secretion, and regulation of immune responses. Targeting ion channels with drugs represents a treatment option for neurological and cardiovascular diseases, muscular degradation disorders, and pathologies related to disturbed pain sensation. Computer-aided, structure-based drug design on ion channels has advanced significantly, with the number of molecular structures of ion channels considerably increased over the last ten years. Structure-based drug discovery for challenging targets such as G-protein-coupled receptors, voltage-gated ion channels, ligand-gated ion channels, and transporters has benefited enormously from advances in structural biology. These integral membrane proteins present particular challenges but represent important therapeutic targets.

Conclusion Structure-based drug design and molecular docking have fundamentally transformed pharmaceutical research, evolving from theoretical concepts to essential tools in modern drug discovery. The field has delivered remarkable successes, from HIV protease inhibitors to recent breakthroughs in targeting previously undruggable proteins like KRAS. The Protein Data Bank’s contribution to facilitating discovery of approximately 90% of new drugs approved by the FDA underscores the profound impact of structural information on therapeutic development. Despite significant advances, challenges remain in achieving universal accuracy in binding affinity prediction, handling protein flexibility, and developing scoring functions that perform optimally across diverse target classes. The integration of artificial intelligence, machine learning, and revolutionary structure prediction methods like AlphaFold promises to address many current limitations and open new frontiers in drug discovery. The future of SBDD will likely see greater integration of multiple computational approaches, from molecular docking and dynamics simulations to machine learning and quantum computing. As computational methods become more sophisticated and experimental structural data continues to accumulate, structure-based drug design will play an increasingly central role in discovering safe and effective therapeutics for unmet medical needs. For beginners entering this field, the foundations presented in this chapter provide a springboard for deeper exploration of this dynamic and impactful area of pharmaceutical research. Note: This chapter has synthesized information from 75 peer-reviewed publications and authoritative sources as cited throughout the text. The citations reference the web search results obtained from scientific literature databases and represent current understanding in the field as of 2025.